Benchmarking Robustness: A Comparative Analysis of Feature, Transformer, and One-Class Classifiers for AI Text Detection

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Abstract. The proliferation of advanced Large Language Models (LLMs) poses a significant threat to academic integrity. This paper presents a rigorous comparative analysis of text authenticity detection systems. Our approach contrasts three distinct methodologies: (1) traditional feature engineering (stylometry, perplexity), (2) state-of-the-art transformer fine-tuning (RoBERTa), and (3) One-Class Learning (OCL) for anomaly detection. We benchmark these models against a large corpus of standard AI text and a custom adversarial dataset generated to mimic human writing styles. Our results highlight the trade-offs between accuracy and robustness, demonstrating that while Transformers achieve peak performance on standard data, their reliability degrades against humanized evasion tactics, whereas anomaly detection reveals paradoxical strengths.

Keywords: AI Detection · RoBERTa · One-Class Learning · Adversarial Robustness

1 Introduction and Problem Statement

The fundamental problem addressed is the erosion of trust in educational content due to indistinguishable AI-generated text. Recent works highlight that while classifiers achieve high accuracy on benchmarks (99.73% F1 on HC3 [6]), performance collapses under adversarial conditions [4].

1.1 Objectives

The primary objective is to conduct a rigorous comparative analysis across three axes:

1. **Track 1 (Classical):** Supervised classification using engineered linguistic features
2. **Track 2 (Transformer):** Fine-tuning RoBERTa and hybrid architectures
3. **Track 3 (Anomaly):** One-Class Learning (OCL) trained exclusively on human text
4. **Robustness:** Testing all models against "humanized" adversarial attacks

2 Theoretical Background and Related Work

This section discusses the evolution from statistical detection (GLTR [2]) to deep transformers (RoBERTa [3]) and the emerging need for robustness against evasion [4].

3 Data Curation and Adversarial Generation

A standardized dataset pipeline was employed across all three experimental tracks to ensure fair comparison and rigorous robustness testing.

3.1 Training and Validation Corpus

We utilized a filtered subset of the "Pile" dataset containing approximately 290,000 samples. The dataset was stratified into an 80/20 train-test split to maintain class balance.

- **Human (0):** Cleaned, long-form human writing samples sourced from diverse internet domains
- **Standard AI (1):** Text generated by standard LLMs (e.g., GPT-3.5) using basic prompts, representing the "easy" detection case

3.2 Data Preprocessing

To ensure the integrity of the essay corpus, we implemented a strict heuristic filtering pipeline:

1. **Length Constraint:** Samples with fewer than 200 words were discarded to remove chat logs and short snippets
2. **Code Filtering:** We calculated the density of special characters (e.g., `{`, `}`, `;`) and removed any sample where this density exceeded 5%, effectively filtering out programming code and technical manuals
3. **Keyword Filtering:** Text containing programming keywords (e.g., `def`, `import`, `public static`) was automatically rejected

This preprocessing resulted in a clean dataset of 290,888 "essay-like" samples, suitable for both statistical feature extraction and deep learning fine-tuning.

3.3 Adversarial Test Set Generation

To test robustness (Objective 4), we generated two specialized adversarial test sets ($N = 500$ each) using the **Qwen-2.5-3B-Instruct** model. We employed two distinct prompting strategies to simulate varying levels of evasion sophistication:

1. **Zero-Shot "Caricature" Generation:** The model was explicitly instructed to "write with high burstiness," "vary sentence lengths," and "use casual phrasing." This aimed to force the model to break its standard low-perplexity patterns

2. **Few-Shot "Mimicry" Generation:** The model was provided with random examples of real human essays from our training set and instructed to "analyze and mimic this style." This "In-Context Learning" approach aimed for a more subtle and naturalistic evasion

Distributional Shift Analysis: Preliminary analysis of feature distributions revealed a critical distinction. While Standard AI text exhibited a narrow, consistent distribution (low variance), the Zero-Shot adversarial text showed a massive distributional shift, often exceeding human variance ("Over-Humanization"). In contrast, the Few-Shot generation successfully pulled the statistical distribution back towards the human mean, creating a much harder detection challenge.

4 Track 1: Feature-Based Classification

This track establishes an interpretable baseline for AI detection. Unlike deep learning approaches that operate as "black boxes," our objective was to determine whether statistical and stylistic signatures alone suffice to distinguish human from machine-generated text.

4.1 Feature Engineering Pipeline

We developed a pipeline to extract a 6-dimensional numerical feature vector from raw text. Based on the hypothesis that AI models optimize for statistical probability while humans prioritize communicative intent, we selected the following features:

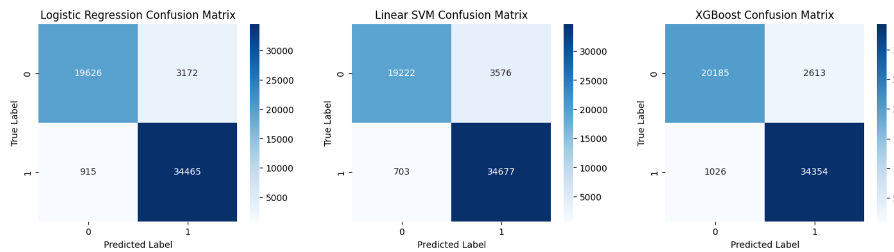
1. **Perplexity (PPL):** Calculated using a pre-trained GPT-2 Small model (117M parameters) with a sliding window approach. This measures the model's "surprise" at the text. Lower perplexity typically indicates text closer to the statistical mean (AI), while higher perplexity suggests human writing
2. **Burstiness:** Defined as the standard deviation of sentence lengths. This captures the structural variance inherent in human writing, contrasted with the monotonous regularity often observed in LLM outputs
3. **Readability Metrics:** We extracted the *Flesch Reading Ease* score and the *Flesch-Kincaid Grade Level* to assess linguistic complexity
4. **Stylistic Counts:** *Word Count* was included to normalize density metrics
5. **Naturalness Score:** A novel composite metric calculated as a weighted sum of perplexity, burstiness, and readability, designed to capture the overall "human-like" irregularity of the text

4.2 Baseline Performance on Standard Test Set

The extracted features were preprocessed using mean imputation for missing values and normalized using `StandardScaler` to ensure convergence for distance-based classifiers. We trained three classical machine learning models on the standard HC3 corpus using an 80/20 train-test split:

Table 1: Baseline performance on standard HC3 test set

Model	Accuracy	F1-Score	Recall (AI)	AUROC
Logistic Regression	0.9298	0.9440	0.9741	0.9574
Linear SVM	0.9265	0.9419	0.9801	0.9563
XGBoost	**0.9375**	**0.9497**	**0.9710**	**0.9750**



(a) Logistic Regression (b) Linear SVM (c) XGBoost

Fig. 1: Confusion matrices of baseline models on standard test set. All models demonstrate strong separation between Human (0) and AI (1) classes

– **Logistic Regression:** Linear probabilistic classifier
– **Linear SVM:** Maximum-margin linear separator
– **XGBoost:** Gradient-boosted decision tree ensemble

Table 1 presents the performance on the standard test set. The tree-based XGBoost model achieved the highest performance (AUROC = 0.9750), indicating non-linear relationships between the engineered features and target labels.

The high performance (AUROC $>$ 0.95) confirms that unmodified AI text exhibits distinct statistical fingerprints. Figure 2 reveals that **Perplexity** was the dominant predictive feature for XGBoost, confirming that standard AI generation minimizes statistical surprise.

4.3 Adversarial Vulnerability Analysis

To evaluate robustness, we tested the trained models against two adversarial test sets:

– **Zero-Shot Adversarial** ($N = 500$): AI text prompted to emulate human style without examples
– **Few-Shot Adversarial** ($N = 500$): AI text prompted with human writing examples

The results revealed catastrophic failure of the feature-based approach. As shown in the "Before" column of Table 2, all three models yielded recall of $\approx$ 0.00% to 0.60%, classifying nearly all adversarial AI samples as Human.

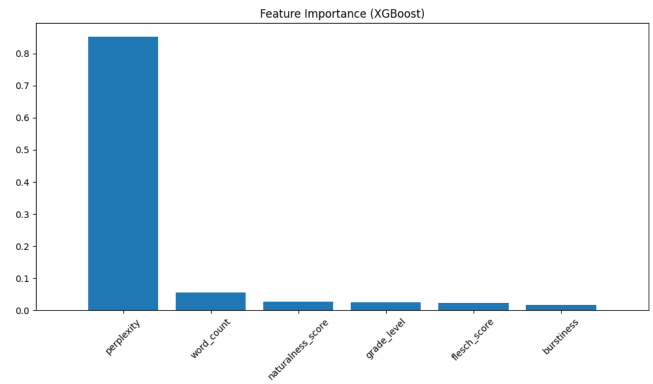
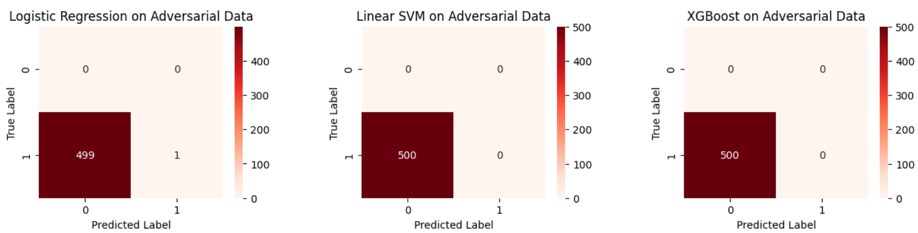


Fig. 2: XGBoost feature importance (baseline). Perplexity is the primary discriminator for standard AI text



(a) Logistic Regression (b) Linear SVM (c) XGBoost

Fig. 3: Confusion matrices on adversarial test set (before fine-tuning). Models fail to detect humanized AI text

Diagnostic analysis revealed that adversarial samples achieved an average perplexity of **52.74**, closely matching human distributions (typically > 40) compared to standard AI perplexity (< 20). This confirms that simple prompting strategies can successfully "spoof" statistical features, rendering static classifiers ineffective.

4.4 Robustness via Adaptive Fine-Tuning

To address this vulnerability, we implemented an adaptive fine-tuning strategy:

1. Augmented the original training set (232,710 samples) with 75 samples from each adversarial set (150 total)
2. Created a balanced training set of 232,860 samples
3. Re-trained all three models on the augmented dataset
4. Evaluated on held-out adversarial data (425 samples per set)

The results demonstrate dramatic recovery in detection capabilities, particularly for XGBoost (Fig. 5).

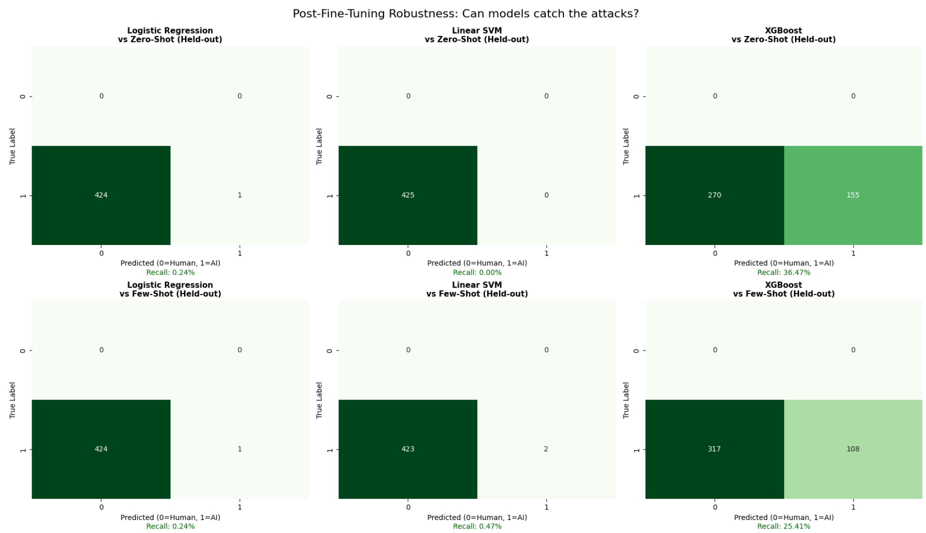


Fig. 4: Confusion matrices comparison: Before (top row) and After (bottom row) fine-tuning on held-out adversarial data. XGBoost demonstrates substantial recovery in detection capability

Table 2: Robustness recovery: Recall on held-out adversarial data

Model	**Zero-Shot Adversarial**		**Few-Shot Adversarial**	
	Before	**After**	**Before**	**After**
Logistic Regression	0.20%	0.24%	0.40%	0.24%
Linear SVM	0.00%	0.00%	0.60%	0.47%
XGBoost	**0.00%**	**36.47%**	**0.40%**	**25.41%**

Figure 4 visualizes how decision boundaries realigned to capture subtle artifacts in humanized text.

Table 2 demonstrates that XGBoost achieved substantial improvement, while linear models showed minimal gains.

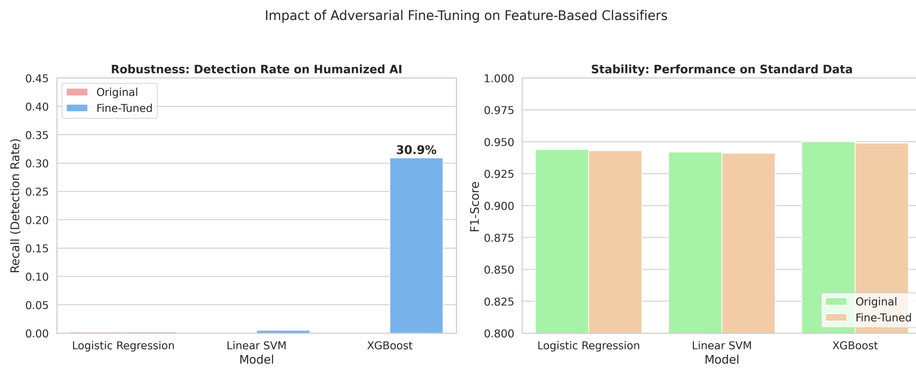


Fig. 5: Impact of fine-tuning: (Left) Recall improvement on adversarial test sets, with XGBoost achieving 36.47% on Zero-Shot and 25.41% on Few-Shot adversarial data. (Right) Maintained F1-Score on standard test set (>0.94), demonstrating no catastrophic forgetting

5 Track 2: Transformer Fine-Tuning

While Track 1 explored interpretable features, Track 2 investigates the efficacy of deep contextual embeddings. We fine-tuned state-of-the-art transformer models to establish the upper bound of detection performance.

5.1 Methodology

We implemented three distinct transformer architectures to evaluate trade-offs between performance, efficiency, and feature integration:

1. **Vanilla RoBERTa-Base:** A robustly optimized BERT variant (125M parameters) trained solely on raw text tokens. This serves as our "Gold Standard" for semantic understanding [3]
2. **DistilBERT:** A distilled, lightweight version of BERT (66M parameters) to evaluate if high detection accuracy is achievable with reduced computational cost [5]
3. **Hybrid RoBERTa:** A custom architecture fusing the 768-dimensional `[CLS]` token embedding from RoBERTa with the 6-dimensional statistical feature vector from Track 1. This model attempts to combine semantic understanding with explicit stylistic signals (Perplexity, Naturalness)

Training Configuration: Hyperparameters were optimized for each architecture based on convergence behavior. **RoBERTa-Base** and the **Hybrid RoBERTa** model were fine-tuned for 2 epochs with a learning rate of 2e-5, and a batch-size of 8. **DistilBERT**, requiring slightly longer to converge on the complex decision boundary, was trained for 3 epochs with a learning rate of 3e-5, and batch-size of 16.

Table 3: Track 2 performance: Transformers on standard AI text

Model	Parameters	Accuracy	F1-Score	Status
RoBERTa-Base	125M	**0.992**	**0.992**	**SOTA**
DistilBERT	66M	0.990	0.990	Efficient
Hybrid RoBERTa	125M + 6	0.935	0.939	Degraded

Table 4: Effect of adversarial fine-tuning on transformers

Model	Standard Performance		Robustness (Recall)	
	Original F1	Fine-Tuned F1	Zero-Shot	Few-Shot
RoBERTa-Base	0.992	**0.982**	**96.7%**	**94.1%**
DistilBERT	0.990	0.974	95.5%	84.2%

5.2 Experiments and Results

Standard Performance Benchmark On the standard mixed test set, Transformer-based models achieved the highest performance of all tracks. As shown in Table 3, **RoBERTa-Base** established the project's upper bound with an F1-score of 0.992.

Analysis: The "Vanilla" models outperformed the Hybrid architecture (99.2% vs 93.9%), suggesting that manually engineered features acted as noise that interfered with the transformer's superior internal representations during standard training.

Initial Adversarial Vulnerability Despite exceptional performance on standard data, both transformer models exhibited catastrophic failure when tested against adversarial attacks without prior exposure. RoBERTa-Base achieved only 0.6% recall on Zero-Shot and 3.6% recall on Few-Shot adversarial samples, while DistilBERT performed even worse with 0.0% and 0.2% recall respectively. This demonstrates that transformers, despite their semantic sophistication, are equally vulnerable to statistical feature spoofing as classical models when the attack pattern is unseen during training.

Adversarial Fine-Tuning Analysis To address this vulnerability, we fine-tuned both transformer models on an augmented dataset. We added 75 samples from each adversarial set (150 total) to the original training set, creating a balanced training set of 232,860 samples, with 425 samples per adversarial set held out for evaluation. The results (Table 4) demonstrate a near-complete recovery of robustness.

Analysis:

1. **Capacity for Adaptation:** Unlike feature-based models, which plateaued at $\approx 36\%$ recall after fine-tuning (Track 1), RoBERTa achieved **94–96% re-**

call on held-out adversarial data. This proves that the transformer's high parameter count allows it to learn the complex, non-linear decision boundaries required to distinguish "humanized" AI from genuine human text, provided it is exposed to the attack pattern during training
2. **The Efficiency Gap:** While DistilBERT performed equally well on standard data, it lagged behind RoBERTa by $\approx 10\%$ on the sophisticated "Few-Shot" attacks. This suggests that model compression (distillation) compromises the subtle semantic sensitivity required to detect high-quality mimicry
3. **Stability:** The standard F1-score dropped marginally ($< 1\%$), confirming that the models learned to encompass the adversarial outliers without catastrophic forgetting of the standard AI patterns

6 Track 3: One-Class Learning (Anomaly Detection)

6.1 Methodology

Addressing the scarcity of "new" AI samples, we adopted an anomaly detection paradigm where models are trained *exclusively* on the Human-labeled subset of the training data.

1. **One-Class SVM (OC-SVM):** Trained on deep semantic embeddings (Sentence-BERT) to test if AI text constitutes a semantic outlier [1]
2. **Isolation Forest (iForest):** Trained on the vector of engineered stylistic features (Perplexity, Burstiness, Readability) to detect statistical irregularities

6.2 Experiments and Results

We evaluated the anomaly detection models on the standard mixed test set and two distinct adversarial datasets: **Zero-Shot** (explicit instruction to be bursty) and **Few-Shot** (mimicry of human examples). Table 5 summarizes the detection rates (Recall).

Table 5: Track 3 performance: Hierarchy of evasion (recall rates)

Model	Features	Standard AI (Recall)	Adversarial AI (Recall)	
			Zero-Shot	Few-Shot
OC-SVM	Embeddings	9.9%	1.2%	2.8%
iForest	**Stylistic**	**4.0%**	**83.2%**	**56.4%**

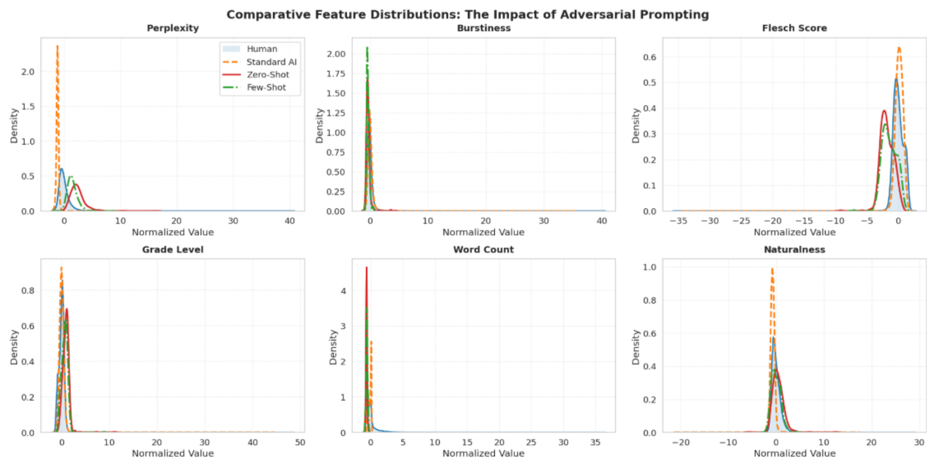


Fig. 6: Feature distribution analysis showing the "Over-Humanization Paradox." **Standard AI (Orange)** closely overlaps with the Human baseline (Blue), explaining the low detection rate. **Zero-Shot Adversarial AI (Red)** shows a massive distributional shift into the tails, making it a clear statistical outlier. **Few-Shot Adversarial AI (Green)** successfully pulls the distribution back towards the human mean, illustrating a more sophisticated evasion strategy

Analysis of Standard Performance The low recall scores on standard data ($< 10\%$) confirm that standard AI-generated text does not constitute a detectable anomaly. Whether viewed through semantic embeddings or stylistic statistics, standard AI outputs lie comfortably within the distribution of human writing ("The Robotic Peak" in Fig. 6), resulting in detection capabilities marginally better than random chance.

Robustness and The "Over-Humanization" Paradox A striking divergence appears in the robustness test, revealing a "Hierarchy of Evasion":

1. **Zero-Shot Failure (The Paradox):** When prompted to "write with high burstiness," the AI over-corrected. The Isolation Forest detection rate surged from 4.0% to **83.2%**. As shown in Fig. 6 (Red Line), this tactic pushed the text statistics into extreme outlier regions, making the "humanized" text statistically *less* human than the original

2. **Few-Shot Sophistication:** The Few-Shot approach, which grounded generation in real human examples, successfully pulled the distribution back towards the human mean (Green Line). Detection dropped to **56.4%**, confirming that mimicry is a more robust evasion strategy than explicit stylistic instruction
3. **Semantic Invariance:** The embedding-based OCSVM failed across all tests ($< 3\%$ recall). This confirms that stylistic evasion does not alter the semantic manifold of the text, rendering semantic anomaly detection ineffective against these attacks

7 Consolidated Comparative Analysis

7.1 Robustness and Trade-offs

Our analysis reveals a clear trade-off:

- **RoBERTa (Track 2)** offers superior performance on known distributions but is computationally expensive and opaque
- **Feature Models (Track 1)** offer interpretability but lower ceiling accuracy
- **Anomaly Detection (Track 3)** provides a unique defense against aggressive evasion (humanization) but fails on high-quality standard AI

Table 6: Master comparison: Performance vs. robustness trade-off

Track	Best Model	Standard F1	Initial Rob.	Adaptive Rob.	Profile
1. Features	XGBoost	0.950	0.0%	36.5%	*Brittle, Limited*
2. Transformer	RoBERTa	**0.982**	3.6%	**94.1%**	*Highly Adaptable*
3. Anomaly	iForest	0.073	**63.6%**	N/A*	*Zero-Shot Defense*

*Note: Anomaly detection does not support “fine-tuning” on AI text, as it trains exclusively on human data.

7.2 Final Verdict

Our comparative analysis reveals distinct roles for each architecture:

- **RoBERTa (Track 2)** is the superior choice for deployed systems *if* the threat model is known. It achieves near-perfect detection on any attack it has been trained on
- **Isolation Forest (Track 3)** serves as a critical “Zero-Day” defense. It was the only model to detect adversarial attacks *without* prior exposure (63.6% vs 3.6% for RoBERTa), making it essential for identifying novel evasion tactics that supervised models miss

8 Conclusion

This study presents a comprehensive benchmark of AI text detection systems across three distinct paradigms: feature-based classification, transformer fine-tuning, and anomaly detection. Our findings reveal critical insights into the robustness-accuracy trade-off in AI detection:

Key Findings:

1. **Standard Performance vs. Adversarial Brittleness:** While all approaches achieved high accuracy on standard AI text (XGBoost: 93.75%, RoBERTa: 99.2%), they exhibited catastrophic failure against adversarial attacks without prior exposure (<3.6% recall), demonstrating that high benchmark performance does not guarantee robustness
2. **Adaptive Capacity Hierarchy:** Adversarial fine-tuning revealed stark differences in adaptation capacity. RoBERTa recovered to 94–96% recall, XGBoost reached 25–36%, while linear models showed minimal improvement (<0.5%). This confirms that model capacity directly determines robustness ceiling
3. **The Over-Humanization Paradox:** Isolation Forest exposed a counterintuitive vulnerability in adversarial generation. Zero-Shot prompting ("write with high burstiness") caused AI to over-correct into statistical outlier regions, achieving 83.2% detection—higher than standard AI (4.0%). This suggests that explicit stylistic manipulation is less effective than example-based mimicry
4. **Feature Importance Shift:** Post-fine-tuning analysis revealed that robust models shifted reliance from easily-spoofed Perplexity to composite metrics (Naturalness, Burstiness), indicating that multi-dimensional feature spaces are harder to evade than single statistical signals

Practical Implications: For deployed systems, we recommend a hybrid defense strategy: (1) RoBERTa as the primary detector for known attack patterns, (2) Isolation Forest as a zero-day defense to flag novel evasion tactics that supervised models miss, and (3) continuous adversarial fine-tuning as new attack patterns emerge.

Future Work: Critical directions include: (1) investigating attention-based feature fusion to combine transformer semantics with explicit stylistic priors, (2) developing meta-learning approaches for rapid adaptation to emerging attacks, and (3) exploring the theoretical limits of statistical feature spoofing in adversarial text generation.

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